End-Term Examination

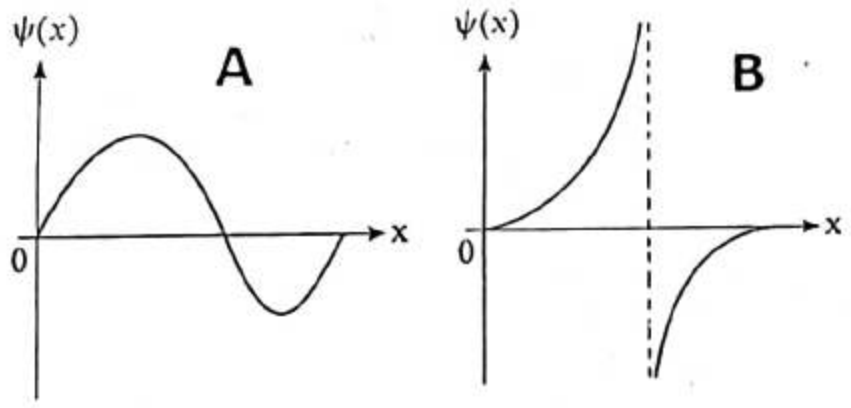
(CBCS)(SUBJECTIVE TYPE) (Offline)

Programme: <B.Tech. (AI&ML/CSE-AI/ECE-AI/ECE/M&C), Semester: <2nd>

(May , 2026)

Subject Code: BAS 102	Subject: Applied Physics
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)
	a) If light of 660 nm wavelength has a spectral width of 1.5×10^9 Hz, what is its coherence length and coherence time?	
	b) How many orders will be observed by a grating having 4000 lines/cm by visible light in the wavelength range of 4000-7000 Å?	
	c) Given a solenoidal vector $\vec{A} = (x + 3y)\,\hat{\imath} + (2y + 3z)\,\hat{\jmath} + (x + az)\,\hat{k}$. Find the value of a.	
	d) The current density is given by $\vec{J} = (x^2\,\hat{\imath} + 2yz\,\hat{\jmath} + z^2\,\hat{k})$ A/m^2. Find the rate of change of charge density at the point (5,0,2).	
	e) State de Broglie Hypothesis. Calculate the de Broglie wavelength of an electron (in Å) if it is accelerated from rest through a potential difference V=100 V.	
	f) Which of the wave functions A and B shown in the figure is/are physically acceptable? Give reason. 	
	g) Make a labelled schematic of optical communication system.	
	h) Is lasing action possible in a two level system? Give reason.	
	UNIT I	
Q2	a. Explain Fraunhoffer diffraction at a single slit and discuss the intensity distribution pattern. b. Give the mathematical analysis of the case when two mutually perpendicular harmonic waves of different amplitudes and phase superimpose to produce a polarized wave.	(5,5)
Q3	a. Derive an expression for intensity at a point P due to Fresnel diffraction at a straight edge. Show the intensity curve for the illuminated and shadow regions. b. What is the phenomenon of double refraction? Give a well labelled diagram of a Nicol prism.	(5,5)

	UNIT II	
Q4	**a.** Explain the failure of Ampere's circuital law and Maxwell's modification of Ampere's circuital law in its generalized form. **b.** State Poynting Theorem. Give the physical significance of Poynting Vector.	**(5,5)**
Q5	Discuss the propagation of plane monochromatic electromagnetic waves in a conducting media. Derive the dispersion equation and thus derive (i) Phase velocity, (ii) impedance (iii) Skin depth in a good conductor.	**(10)**
	UNIT III	
Q6	**a.**Give the properties of the quantum mechanical wave function. Find the normalization constant of $\psi(x)=A(1+\sin x)$, in the range $0 \le x \le \pi$ **b.** State Heisenberg Uncertainity principle. Show that an electron cannot reside within the nucleus of an atom.	**a.(3,3)** **b. 4**
Q7	**a.** Derive the time independent Schrodinger Wave equation and give its general solution. **b.**Find the ground state energy of the electron confined in a 1-D infinite potential well of length 1 nm.	**(5,5)**
	UNIT IV	
Q8	**a.** Discuss Einstein's transition probabilities and derive the relation between Einstein's A and B coefficients with due discussion on the results. **b.** Explain the construction and working of He-Ne Laser with energy diagram.	**(5,5)**
Q9	**a.** Differentiate between a step index and graded index fiber. The refractive index of the core and cladding of a fiber is 1.52 and 1.41 respectively. Calculate critical angle and maximum incidence angle in degrees. **b.** How many modes can propagate in a step index fiber with a core diameter 40 μm, if the refractive indices of the core and cladding are 1.461 and 1.456 respectively and the wavelength of light is 8500 Å.	**(7,3)**